Lesson 4 LT 4.3 · Why the mRNA matches the coding strand

Work with a partner and the base-pair cards. DNA cards are square. RNA cards are round.

1. Build a double strand. Lay out the top strand below in DNA cards. Under it, lay the base that pairs with each one. Write the bottom strand.

Top strand

A	T	G	C	C	G	T	T	A

Bottom strand

2. Unzip it. Slide the two strands apart. The cell reads the **bottom** strand. That one is the **template strand**. The top one is the **coding strand**.

3. Transcribe. Lay RNA cards along the template, base by base. Remember: A on the DNA pairs with **U** in RNA.

mRNA

4. Compare. Lift your mRNA row and set it over the top strand. What do you notice?

5. Explain. Why does the mRNA match the coding strand, except for one base? Use: *template strand*, *complementary*, *coding strand*, *uracil*.

Reading the codon table

Example: GCA. First base **G**: find the G row on the left. Second base **C**: find the C column across the top. Third base **A**: find A on the right side of that box. GCA = **ala** (alanine).

mRNA codon	Amino acid
UUU	
CAG	
AGA	
GGC	
UAA	
AUG	
CCU	
GAU	

Which codon codes for **tryptophan** (trp)? _____

Which **two** codons code for **lysine** (lys)? _____ _____

Why can a changed base sometimes keep the same amino acid? Look at the four codons for alanine.

_____

_____

Lesson 6 LT 4.3 · One changed base

- A **mutation** is a change in the sequence of DNA. Mutations happen by chance: a copying mistake during replication, or damage from things like UV light.
- You find a mutation by **comparing** a sequence with an unchanged copy.
- When **one base is swapped** (a point mutation), the codon changes. Read it with the codon table. The amino acid may **stay the same**, **change to a different one**, or become a **STOP**, so the protein ends early.
- When a base is **added or removed**, every codon after it shifts. This is a **frameshift**.
- A mutation can be harmful, helpful, or have no effect. It can be passed on even if it does not show.

Work it. Transcribe, then translate with the codon table on p. 11.

Original DNA	**TAC**	**GGA**	**CTT**	**ACC**	**ATT**
mRNA					
Amino acid					

Changed DNA	**TAC**	**GGA**	**CTT**	**ACT**	**ATT**
mRNA					
Amino acid					

1. Circle the base that changed. What happened to the protein?

2. Now go back to the original and change the **C** in **CTT** to a **T**. Which kind of change is it: same amino acid, different amino acid, or early stop?